

USER INFORMATION READ CAREFULLY

HEPARINOL® 1000 IU/mL (Heparin Sodium (Bovine) 1000 IU/mL Injection BP)

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Composition:

Each HEPARINOL® 1000 IU/mL contains Heparin Sodium (Bovine) 1000 IU/mL and Benzyl Alcohol 1% w/v.
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Pharmacodynamics:

Heparin is an anticoagulant which prevent the coagulant of blood in vivo or in vitro. The whole blood clotting time, the thrombin time and the one stage prothrombin time are prolonged following therapeutic doses of Heparin. After injection, Heparin is extensively bound to Plasma Proteins. It does not cross the placenta or appear in the milk of nursing mothers.

A heterogeneous group of mucopolysaccharides derived from the mast cells of animal tissues. It binds with antithrombin III, accelerating the rate at which antithrombin III neutralizes activated forms of factors XII, XI, IX, VII and II.

Pharmacokinetics:

Heparin is not absorbed through the gastrointestinal tract and must be administered parenterally. Its onset of action is immediately following I.V administration. There may be considerable variation among patients in the extent of absorption following deep subcutaneous injection of heparin; however, the onset of activity usually occurs within 20-60 minutes.

The metabolic fate of heparin is not fully understood. No biotransformation in plasma or liver, nor any renal excretory mechanism has been identified as primarily responsible for elimination of the drug. It has been suggested that transfer and storage in the reticuloendothelial system may play a role, or that heparin may be partially metabolised in the liver. After administration of large doses intravenously, a small fraction of unchanged drug is excreted in the urine.

Indication:

Heparin is indicated for prophylaxis and treatment of thrombo-embolic disorders such as thrombophlebitis, pulmonary embolism, myocardial infarction, arterial embolism and other occlusive vascular disease. It is also used to prevent thrombo-embolic complication from cardiac and vascular surgery, dialysis, frostbite and other perfusion procedures. Heparin is also used as anticoagulant in blood transfusions and for laboratory purposes.

Recommended Dosage:

Express dose in units only; dose must be individually titrated to desired effect (usually 1.5 - 2.5 times control clotting test used).

1. Treatment of venous thrombosis or pulmonary embolus. Duration of therapy is 7-10 days, then followed by oral anticoagulation.
 - i. Continuous IV solution: 50 - 100 units/kg initially; then 10-15 units/kg/hr (venous thrombosis) or 20 units/kg/hr (pulmonary embolus).
 - ii. Intermittent IV: 75-125 units/kg q 4 hr.
 - iii. Subcutaneous : 10000 - 20000 units initially (preceded by a 5000 units IV loading dose), then 8000 - 10000 units q 8 hr or 15000 - 20000 units q 12 hr.
2. SC for prophylaxis of deep vein thrombosis (low-dose) 5000 units 2 hr before surgery, repeated q 8-12 hr until patient is ambulatory.
3. IV for heparin lock flush inject sufficient solution (of 10 - 100 units/ml) into injection hub to fill the entire after each heparin lock use.
4. Haemodialysis : 7,500-12,500 IU is normally required per dialysis.
5. Myocardial infarction : 5,000 IU s.c. every 12 hours beginning during the 12 hours following the first sign of myocardial infarction.

Route of Administration: IV/SC

Contraindications:

Heparin therapy is contraindicated in patients known to have hypersensitivity reaction, any uncontrolled active bleeding stage; and thrombocytopenia. It should not be used in the presence of actual or potential haemorrhagic stages, e.g. haemophilia, ascorbic acid deficiency or increased capillary fragility, threatened abortion, immediate postpartum period, subacute bacterial endocarditis, severe hypertension, suspected intracranial haemorrhage, gastric or duodenal ulcers, advanced renal or hepatic disease and after eye, brain or spinal cord surgery.

Warnings and Precautions:

Warnings

HEPARINOL® contains benzyl alcohol, its use should be avoided in children below 2 years old and should not be used in neonates.

Precautions

Heparin therapy should be monitored carefully to reduce the risk of overdosage, haemorrhage and serious adverse reactions such as delayed onset thrombocytopenia.

Risk factors for haemorrhage may include IM injections, trauma, recent surgery, women over 60 years, malignancy, peptic ulcer disease, potential bleeding sites and acquired or congenital homeostatic defects.

As heparin is derived from animal tissue (bovine mucous), it should be used with caution in patients with a history of allergy. Before a therapeutic dose is given to such a patient, a trial dose of 1,000 units may be advisable.

Interactions with Other Medicaments:

Aspirin effects on platelet adhesiveness might leave the heparin treated patient more prone haemorrhage. Non steroidal anti-inflammatory agents, dextran, dipyridamole and thrombolytic agents such as streptokinase or urokinase may also increase the risk of haemorrhage.

Statement on Usage During Pregnancy and Lactation:**Pregnancy**

Category C (FDA)

Safety for use during pregnancy has not been established. Use only when clearly needed and when the potential benefits outweigh the potential hazards to the fetus. Osteoporosis and thrombocytopenia are the usual risks for mother. Although heparin does not cause malformations, an increased incidence of human foetal loss and prematurity associated with haemorrhage has been reported.

Caution is recommended when heparin is used during the last trimester of pregnancy or during the postpartum period because of the increased risk of maternal bleeding.

Lactation

Heparin is not distributed in milk.

Adverse Effects / Undesirable Effects:

Haemorrhage (bleeding) is a major risk of heparin therapy and may range from minor local ecchymoses to major haemorrhagic complications. Skin necrosis, transient alopecia, diarrhoea and allergic reactions to heparin occur rarely. Delayed onset thrombocytopenia is also a possible complication of therapy. Osteoporosis may occur with doses of 15,000 units/day or more for 6 months or longer.

Overdose and Treatment:

Slight haemorrhage due to overdosage can usually be treated by withdrawing the drug. Severe bleeding may be reduced by the administration of protamine sulfate. Protamine sulfate should be administered intravenously. The dose of protamine required to antagonise heparin can be estimated; 1mg of protamine should be administered for every 100 units of heparin remaining in the patient.

Alternatively the requirement for protamine can be determined directly in vitro by titration of the patient's blood with the protein. To avoid circulatory side effects, injection protamine should be given slowly at rate of 5mL/10 minutes and maximum dose at one time in 50 mg.

Parameter to monitor:

Baseline values for APPT or ACT, PT, haematocrit and platelet count. Obtain APPT or ACT (Therapeutic range 1.5-2.5 times control) 3-4 times (or until therapeutic range is achieved) on 1 day, and at least daily thereafter. Platelets and haematocrit every other day; sign of bleeding (melena, haematuria, ecchymosis, haematemesis, epistaxis) daily.

Incompatibilities:

Heparin is incompatible with certain substances in aqueous solution. The following incompatibilities have been reported: hydrocortisone, hyaluronidase, hydroxyzine, some antihistamine, narcotic analgesics, phenothiazines and antibiotics.

Storage Conditions:

Do not store above 30°C. Do not refrigerate.

Shelf life:

3 years from manufacturing date in the proposed storage condition. Do not use after expiry date.

Following the withdrawal of the first dose the remainder should be used within 28 days. This claim is valid under the following

1. Aseptic technique is employed during removal of heparin from the vial.
2. The withdrawal of the Heparin solution is via the rubber bung.
3. Do not refrigerate after opening.

KEEP OUT OF REACH OF CHILDREN**Dosage form and packaging available:**

HEPARINOL® 1000 IU/mL

5mL X 10 vials per box

30mL X 20 vials per box

HEPARINOL® 5000 IU/mL

5mL X 10 vials per box

Manufacturer / Product Registration Holder:

AIN MEDICARE SDN. BHD.

Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA

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